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Distribution System Planning

5.1 Introduction

In a simple sense, distribution system planning means being prepared to meet the electricity needs of the customers in the future. Since the loads change continuously due to changes in population, technology, and in habits of people, planning for electricity needs becomes a very complex and capital-intensive process. It requires the consideration of several objectives simultaneously while also meeting the technical constraints. The overall goal is to minimize the total cost while ensuring that the system has adequate capacity to supply the load for the future with adequate reliability and acceptable voltage quality. Good plans should be able to address the short-term needs of one to two years as well as the long-term needs ranging from 5 to 10 years.

The first stage in planning is to forecast the load of the future. In addition to knowing the load growth, it is important to know where the load will be growing. After the areas of load growth and future loads are known, the next step is the location and capacity of substations. The number and size of transformers need to be included. The planning results may also include reinforcing the existing substations as well as building new substations. The final step is the feeder design to deliver power from the substations to the customers. Feeder design includes both the primary and the secondary systems. The decisions include the number of primary feeders, the size of conductors and their routing.

At the secondary level, the decisions include the location and size of the distribution transformers. Most of the secondary construction in the United States has been overhead so far. However, now there is a trend toward making new secondary construction underground. Although the underground secondary system can be 5–10 times more expensive than the overhead secondary system, the underground secondary system is preferred by many localities for greater reliability and aesthetics.

5.2 Traditional vs. Modern Approaches to Planning

Legacy distribution systems have very little metering information available from the system. Usually, only loads on the substation transformers and the feeders are measured. Thus, utilities have very little information beyond the feeder. Utilities typically rely on customer billing data and the total installed distribution transformer capacity to get an estimate of the existing load. As part of the load research activity, utilities also installed recording devices at selected customer locations to record loads at a predetermined interval (5 minutes, 15 minutes, 30 minutes, or 1 hour) to get an idea of the daily load profiles for different classes of customers. These load profiles are also useful for planning purposes. However, with limited information available from the system, distribution planning was ad hoc, which would often result in oversized systems.

Modern distribution systems have significantly more metering capabilities, both at the system level due to distribution automation and at the customer level due to advanced metering infrastructure. Although many advances in automation and metering have been made over the years, by and large the existing distribution systems around the world do not have much metering capabilities. As more systems become automated, more accurate data from different parts of the system become available to the distribution planners. Such availability of data removes the need to make assumptions about various factors about which no data or information were available.

Traditionally, the distribution systems are designed to meet peak demand. The modern approach, based on risk analysis, requires the number of hours spent at different load levels. These data allow the planners to determine the risk of insufficient capacity to meet the demand over a period of time. Specifically, the number of hours under peak conditions is very important for risk assessment. Modern distribution planning should consider several other factors, such as system reliability, resiliency, aging assets, equipment loading close to margin, and regulatory environment in the planning process. Including details of all these issues is beyond the scope of this chapter. We will give details on only a few of the important factors related to distribution system planning.

5.3 Long-term Load Forecasting

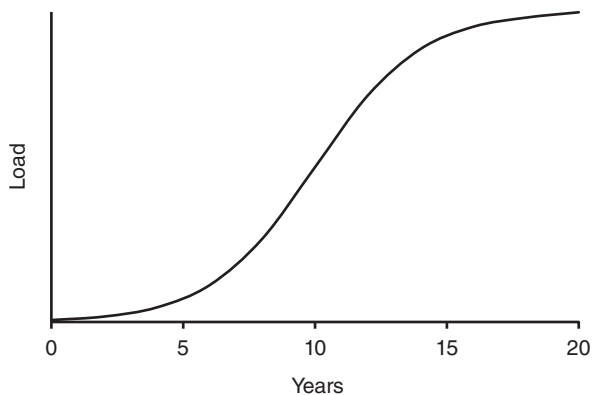
Load forecasting entails determining load in a given area at a future time, which can be from seconds to years. Short-term load forecasts, such as hour ahead or day ahead, are needed for system operation. In contrast, long-term forecasts of loads one year or more into the future are needed for system planning. Since this chapter deals with planning, we only focus on long-term forecasting. Knowledge of future

loads is the most important thing for a planner to know. Based on this knowledge the planner can design facilities in the distribution system to meet these loads with a certain level of reliability. Specifically, the planners need to know the peak load and the time of the peak load. In addition to temporal forecasts, spatial forecasts are important because they tell the locations of load growth in addition to the quantity of load growth. Spatial forecasts allow utilities to focus on areas within their territory where there is a need for new facilities.

Load forecasts depend on the local as well as national trends. For example, every city has its forecast of population growth for the future years. It also has knowledge of new industrial or commercial facilities planned for their localities. At the national level, there may be a push to encourage sales of high-efficiency appliances for energy conservation or a promotion to use light-emitting diode (LED) lamps for lighting. Appliance manufacturers and consumer organizations conduct regular surveys to keep track of consumer behaviors to determine these trends. A planners' job is to use information from such surveys and from local sources to create load forecasts. Past trends in actual load measurements within the distribution system are good indicators of future loads. Another approach to forecasting is based on end-use modeling in which loads of different end uses, such as lighting, water heater, and air conditioner, for different customers are added in a hierarchical manner to obtain the composite load.

If we examine the composite load over the years for a utility's service area, we will most likely find a gradual increase in load. However, for small areas, this model is not very appropriate. As discussed in Ref. [1], load growth in a small area follows an S curve as shown in Figure 5.1. Three parts of this curve are the dormant period, the growth period, and the saturation period. The dormant period typically has very little or no load growth. The growth period is when the load rises rapidly, and the saturation period is when the area is fully built, with very little new construction. The load may continue to grow slightly in the saturation

Figure 5.1 S-curve showing load growth in a small area.



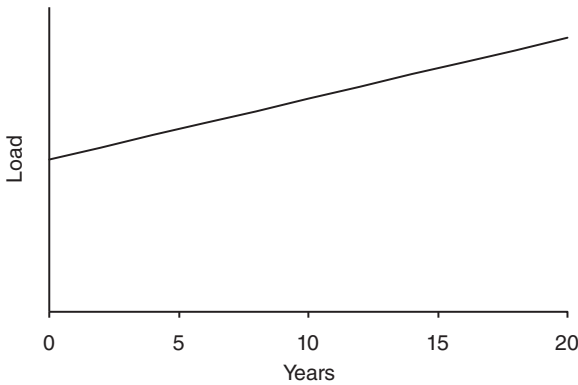


Figure 5.2 Load growth of a service area over the years.

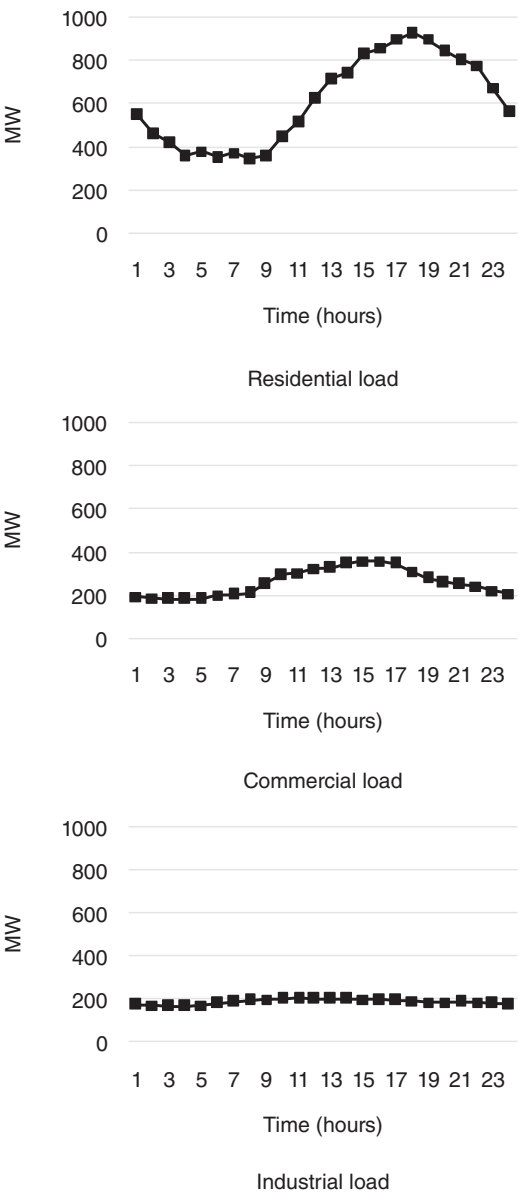
period due to lifestyle changes of the customers in the area. Since all small areas have different S curves, both in terms of load and timing of load, when we combine all the small areas within a utility, the following happens: the composite load shows the linear characteristics with a certain slope showing load growth for a service area as shown in Figure 5.2. Breaking up utility service area into smaller areas is particularly useful for spatial load forecasting.

5.4 Load Characteristics

5.4.1 Customer Classes

Utilities serve customers of different classes, which include residential, commercial, industrial, and agricultural customers. While load demands of customers in different classes differ from one another significantly due to different electricity usage patterns, load demands of customers within a certain class are different, but they follow a similar pattern. For example, load demands of residential customers are low in the night and start to increase in the morning when people start waking up. The load demand continues to rise during the day reaching a peak and then starts to decline in the evening hours. Similarly, commercial and industrial load demands are high during the day but low during the night. Variation in the commercial load demand from day to night is larger than that of industrial loads because several industries tend to operate throughout the 24 hours of the day. Agricultural loads have very different demand patterns, and they depend on the irrigation needs of different crops. The load demands in each class also fluctuate with season. Utilities maintain a database of load demands in each class for planning purposes. Examples of typical load demand characteristics of customers of different classes for a utility are shown in Figure 5.3.

Figure 5.3 Typical aggregate load demands of residential, commercial, and industrial classes at a utility on a summer day.



5.4.2 Loads in a Modern House

Loads in houses today are very different from those 25 years ago. In addition to more efficient appliances, modern homes have new types of loads. Principal among them are computer and entertainment devices. Lighting loads for houses went through a transformation due to replacement of incandescent lamps with compact fluorescent lamps (CFLs) and now LED lamps, which consume substantially lower energy than the replaced ones.

According to a survey done by Energy Information Administration (EIA) for 2015, the average monthly electric energy consumption of 129,811,718 residential customers in the United States was 901 kWh, with an average monthly bill of \$114.03 and an average price of \$0.1265/kWh. Residents of the state of Washington enjoyed the lowest price for electricity at \$0.0909/kWh, while the residents of the state of Hawaii paid the highest price for electricity at \$0.296/kWh. In the lower 48 states, the state of Connecticut had the highest price of electricity at \$0.2094/kWh. The average monthly electricity consumption was the highest in Louisiana at 1286 kWh, and it was the lowest in Maine at 556 kWh. The hot and humid climate of Louisiana required more electricity for air conditioning. Maine has cool summers but brutal winters. Since heating in winter is mainly based on gas, it did not impact the electricity consumption of Maine residents.

The EIA survey also reports that in 2015 electricity contributed to 47%, and natural gas contributed to 44% of energy consumed in homes. Heating, cooling, and ventilation accounted for about half of the home energy consumption. Table 5.1 shows the energy consumed by different usages. Air conditioning used the largest amount of energy, which was followed by space cooling and water

Table 5.1 Energy usage in the US homes in 2015 [2].

Usage	Percentage of energy used
Air conditioning	17
Space heating	15
Water heating	14
Lighting	10
Refrigerators	7
Television and related equipment	7
Clothes dryer	5
Miscellaneous	12
Other	13

Source: Data from U.S. Energy Information Administration [2].

heating. For homes with natural gas-based heating and water heating, space cooling or air conditioning is the largest electrical load with ratings ranging from 1 to 6 kW for a window unit to a large central air conditioner. Lighting consumed 10%, refrigerators 7%, televisions (TVs) and associated equipment 7%, and clothes dryers consumed 5%. Freezers, dishwashers, clothes washers, cooking ranges, microwaves, air handlers, fans, pool/hot tub pumps, hot tub heaters, humidifiers, dehumidifiers, and evaporative coolers consumed 12% of electricity (listed as miscellaneous in the table). The remaining 13% was consumed by computers, home audio, coffee makers, and miscellaneous appliances (listed as other in the table). Of these loads, the refrigerator is rated about 600 W, and it cycles on and off throughout the day to maintain the specified temperature inside it. Some homes have an additional stand-alone freezer, which has a typical rating of 500 W. The clothes washer is rated about 500 W, the dishwasher is about 1.2 kW, and the clothes dryer is about 4 kW. Several appliances are used a few times in a day or once in a few days, depending on the size of the family. Over the years, the electronics portion has gradually increased due to the popularity of large flat screen TVs and computers. On the other hand, the portion for lighting has been decreasing due to the proliferation of LED lighting.

Energy efficiency is a major focus of the government, the industry, and, now, the consumers. In the future, we should expect the electrical rating and energy consumption of different household loads to go down as the appliances, as well as the houses, become more efficient.

5.4.3 Time Aggregation

Instantaneous power, which is obtained by multiplying voltage and current waveforms, shows large fluctuations and is not very useful for planning. Instead, demand (in kW), which is the average power over a selected time period, is used. Typical time periods used by the utility are 1 minute, 5 minutes, 15 minutes, 30 minutes, and 1 hour. Thus, to determine demand for a time period, energy consumed (kWh) over that time period is divided by that time period in hours. As an example, consider that a load consumes 2 kWh of energy in a given 15-minute (0.25 h) period. Therefore, demand for this period is $2/0.25 = 8$ kW. Similarly, if the same load consumes 7.5 kWh of energy over a one-hour period, the demand for this period will be $7.5/1 = 7.5$ kW. Although the demand is over a period of time, the demand reported at a given time is the average load for the time period that ended at the time reported. So, if we are looking at demand at 2:00 a.m. reported on one-hour basis, the reported value at 2:00 a.m. is the average load between 1:00 a.m. and 2:00 a.m. Similarly, if the demand reported duration is 15 minutes, the reported value at 2:00 a.m. is the average load between 1:45 a.m. and 2:00 a.m. Similar logic can be used for other time durations for recording demands.

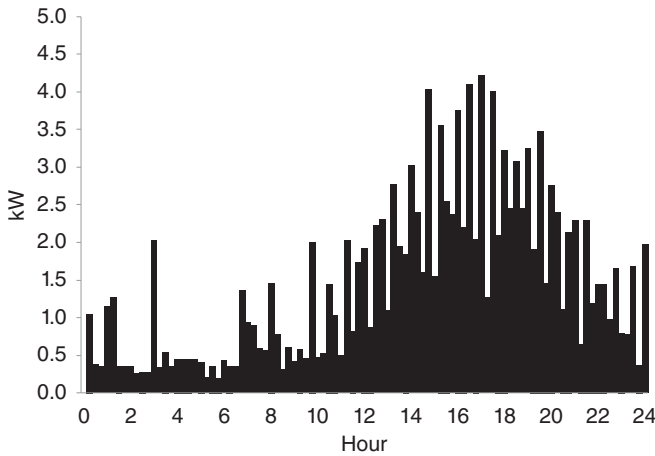


Figure 5.4 Demand of a house on 15-minute basis.

The demand characteristics of a single house based on a small time period have large fluctuations as shown in Figure 5.4. The main reason for these fluctuations is random switching on and off of loads within a house either automatically or by its occupants. Air conditioners, central heating, refrigerators, and water heaters are some loads that turn on and off automatically. On the other hand, lighting, cooking appliances, washer, dryer, computers, and entertainment loads are turned on and off by people. Even while on, some loads may require different amount of power at different times.

The load demand characteristics start to become smoother as we increase the time period. Figures 5.5 and 5.6 show the load demand characteristics of the same house on 30-minutes and 1-hour basis. Usually, time interval larger than one hour is not used to determine the load demand characteristics for planning purposes. Notice that the time aggregation reduces the peak value of the load demand.

5.4.4 Diversity and Coincidence

We noticed in the previous section that temporal aggregation reduces the peak demand of a house by averaging the demand over the selected time duration. Similarly, if we add demands of several houses, we find that the load demand characteristics become smoother as shown in Figures 5.7–5.10. This happens because loads in one house are going on and off at different times than at another house. In utility terminology, this phenomenon is called *load diversity* (LD). Another thing to keep in mind is that all houses have their peak demand at different times. Therefore, the average demand of a group of houses at any time, when any of the houses has its own peak, is lower than the peak demand of that house. For planning purposes,

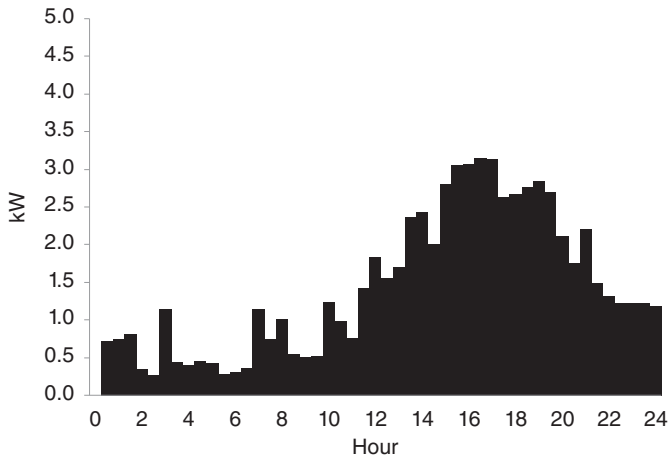


Figure 5.5 Demand of a house on 30-minute basis.

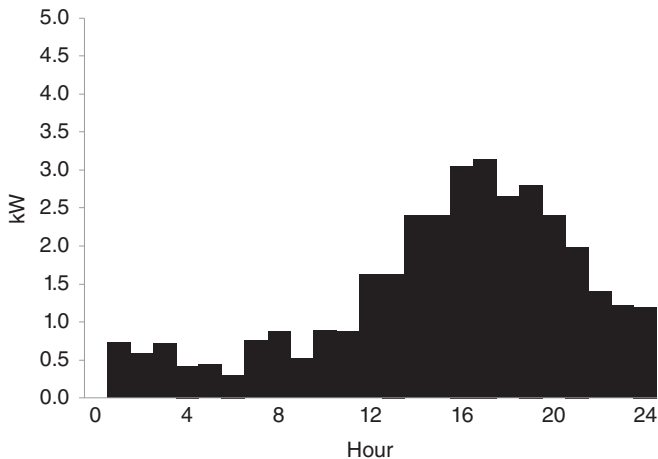


Figure 5.6 Demand of a house on one-hour basis.

the maximum value of the combined demand or *coincident peak demand* over a period of time for a group of loads is of interest. So when we decide the size of the conductor for service to a house, we use the peak value of the combined demand of the devices in the house on the hottest day (or the coldest day in winter-peaking regions). However, when the utility decides on the size of the transformer for supplying power to a house, they look at how many more houses are being served by the same transformer. Since the actual demands of the houses cannot be known in advance, the typical load demand characteristics of houses of a given type for

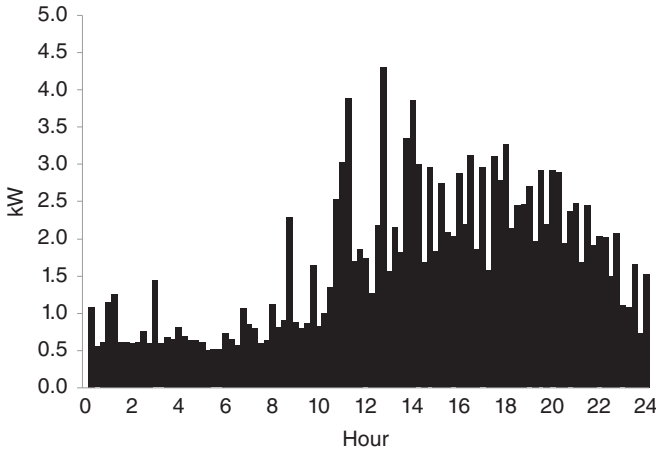


Figure 5.7 Fifteen-minute average load of two houses.

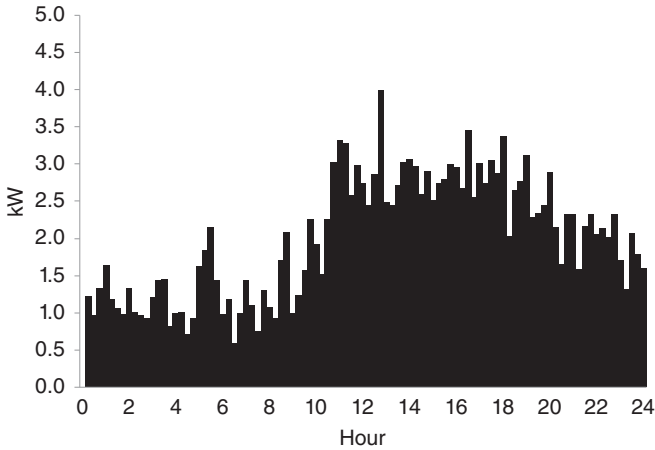


Figure 5.8 Fifteen-minute average load of five houses.

the selected housing subdivision are used in the design. The maximum loads of houses in the group are aggregated and adjusted by a factor called *coincidence factor* to get the coincident peak demand of the group. Let D_{mi} be the peak demand of the i th load in a group of n loads, and D_{mG} be the coincident peak demand of the group. Then, we define the *coincidence factor* (C) as follows:

$$C = \frac{D_{mG}}{\sum_{i=1}^n D_{mi}} \quad (5.1)$$

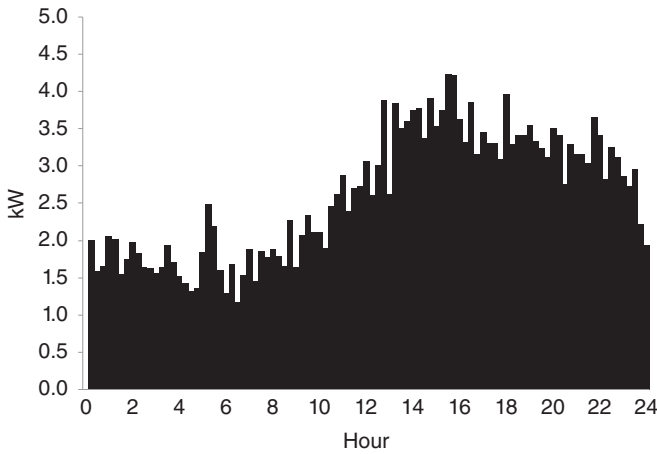


Figure 5.9 Fifteen-minute average load of 10 houses.

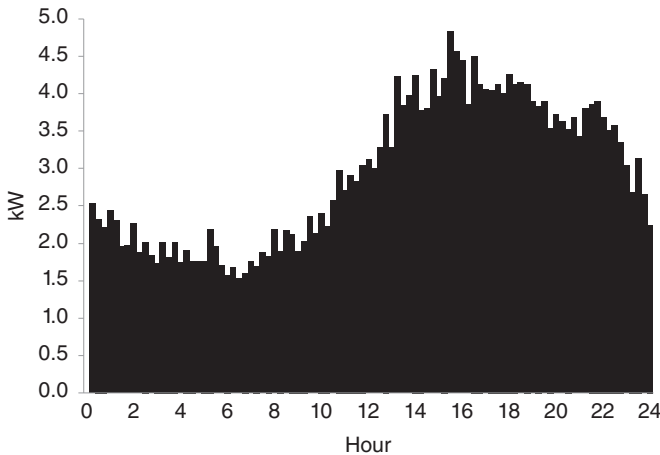


Figure 5.10 Fifteen-minute average load of 20 houses.

The value of the coincidence factor ranges from 0 to 1, and it is a function of the number of houses in a group with its value decreasing with increase in the number of houses. It is not unusual to find the coincidence factor in 0.3–0.6 range for a large group of houses. A graph similar to the one shown in Figure 5.11 can be obtained showing variation in the coincidence factor as a function of n , which is the number of houses in the group. The value of the coincidence factor decreases rapidly for lower values of n but saturates after a certain value. Typically, saturation takes place with n in 15–20 range.

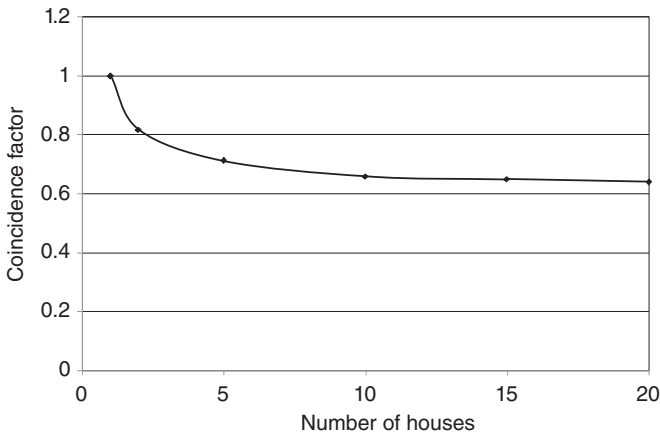


Figure 5.11 Coincidence factor as a function of number of houses.

The inverse of the coincidence factor is defined as the *diversity factor* (D), and it gives an indication of the separation in individual house's peak demands in comparison to the peak demand of the group. Therefore,

$$D = \frac{1}{C} \quad (5.2)$$

Further, the difference between the sum of the individual peak demands and the coincident peak demand of the group is defined as LD, or

$$LD = \sum_{i=1}^n D_{mi} - D_{mG} \quad (5.3)$$

The concept of coincidence can also be applied to loads of different types. For example, if a feeder has to serve residential, commercial, and industrial loads with given load demand characteristics, these characteristics can be used to find the coincident peak demand. This and other factors are then used to find the size of the feeder that would be adequate to serve the combined load.

Example

Find the transformer size needed to serve a group of four houses with average peak load of 5 kW at 0.85 power factor for each house.

Solution

We use Figure 5.10 to find C for $n = 4$, which is 0.73.

Therefore, the coincident peak load of the group = $0.73 \times 5 \times 4 = 14.6$ kW

$$\begin{aligned} \text{kVA at peak load} &= \text{kW at peak} / \text{power factor} \\ &= 14.6 / 0.85 = 17.18 \text{ kVA} \end{aligned}$$

Since this is not a standard size, we look for the next higher standard size, which is 20 kVA.

5.4.5 Demand Factor

The ratio of the maximum load demand over a period to the connected load for any load or class of load is defined as the *demand factor (DF)*.

$$DF = \frac{\text{Maximum Load Demand}}{\text{Connected Load}} \quad (5.4)$$

For example, a house has many loads, but they are not used simultaneously. Only a small portion of them is used simultaneously. Connected load is the total sum of the rated values of all the loads. If we count the ratings of all the light bulbs in a house, we might get a value close to 2 kW. However, if we count the rated values of the lights that are on simultaneously at a time when the maximum number of lights are on, we might get a value of 600 W. Hence, the DF of the lighting load in the given house will be $0.6/2 = 0.3$. Similarly, if we have a 6-kW air conditioner in a house and on the hottest day in summer, it runs for a total of 30 min, the average demand for the hour will be 3 kW. Therefore, the air conditioner's DF is $3/6 = 0.5$. The concept of the DF can be used for individual loads or a group of loads.

5.4.6 Load Duration Curve

So far we have seen that the load in a distribution system changes with time. For short durations such as a day, examination of hourly loads gives useful information related to the highest and the lowest loads with their time of occurrence for the day and pattern of fluctuation of the load during the day. However, if we have to examine the load for a longer duration, such as a year, examination of hourly load for each day of the year becomes a very tedious task. To simplify this task, we condense the load information into *load duration curve*. Load duration curve is a graph of load vs. time for which the load is higher than the specified load. We can easily obtain the two extreme ends of the load duration curve by looking at the highest and the lowest values of the load over a period of time. Since load is never higher than the peak load (L_p), it is higher than L_p for 0% of the time. However, it is always higher than or equal to the lowest load (L_l), or it is higher than or equal to L_l for 100% of the time. Note that we have made an exception for the lowest load for which we are considering greater than equal to instead of greater than the specified load. In between points can be obtained by quantizing the load in equal steps. For example, if the peak load is 17 MW, we could use steps of 2 MW. Too large steps will not give enough data points, and too small steps will give too many data points. Judgment of the user is important in determining the step size.

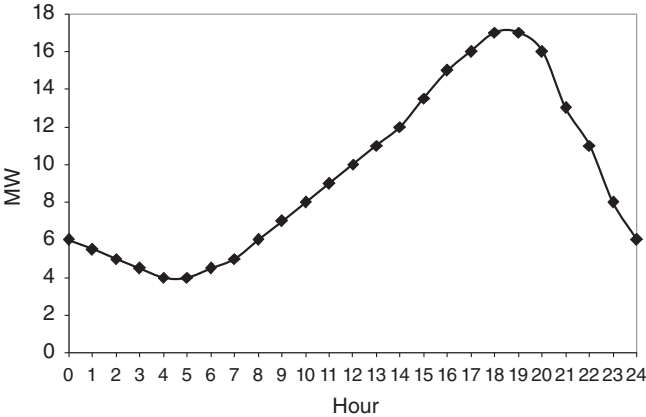


Figure 5.12 Hourly load at a substation. The load shown at a given time is the average load for the hour ending at that time.

To illustrate the idea, let us consider the load characteristics of a load on a substation transformer for a given day, as shown in Figure 5.12. Note that this figure is not based on real data, but this example is created to illustrate the concepts. From this figure, we see that $L_p = 17$ MW and $L_l = 4$ MW. For illustration, let us consider steps of 2 MW, and from the figure count the number of hours for which the load is higher than the selected load. For example, the load is higher than 10 MW for 10 h. Similar counting will give us the data shown in Table 5.2, and the corresponding load duration curve is shown in Figure 5.13. Note that the load at hour zero is not included in this calculation. Load duration curves, especially representing a year, are very important for planning studies. They are used to plan for equipment needed in the system and to estimate losses.

Table 5.2 Data for load duration curve obtained from Figure 5.12.

Load (MW)	Number of hours for which load is higher than this value	% of time for which load is higher than this value
17	0	0
16	2	8.33
14	5	20.83
12	7	29.17
10	10	41.67
8	12	50.00
6	15	62.50
4	24	100.00

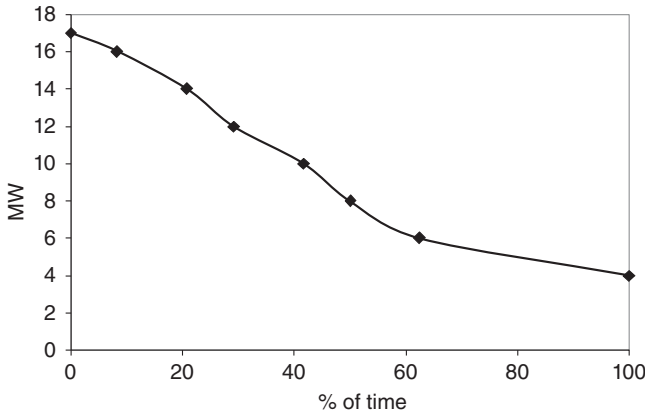


Figure 5.13 Load duration curve corresponding to the load characteristics of Figure 5.12.

5.4.7 Load Factor

Load factor is a term that relates to changes in load over a period of time with respect to the peak load. In other words, it describes the extent to which the load stays close to the peak load. A high load factor implies that the load stays close to the peak load for a large part of the time duration. Mathematically, it is the average load over a period of time divided by the peak load within that time duration, or

$$LF = \frac{L_a}{L_p} = \frac{\left[\frac{\sum_{i=0}^n L_i}{N} \right]}{L_p} \quad (5.5)$$

where L_i is the hourly load at hour i , and N is the total number of hours in the time period under consideration. Notice that the numerator of Eq. (5.5) is equal to the total energy consumed during N hours. For the load characteristics shown in Figure 5.12, with $L_p = 17$ MW and $L_a = 9.5$ MW, we get a load factor of 0.527.

5.4.8 Loss Factor

Similar to the load factor, loss factor (LsF) is the ratio of the average hourly losses to peak hourly losses in the system. LsF is very useful for computing the total losses for a period of time if losses at the peak load are known. For load characteristics in which peak load exists for a very short duration, an approximate value for the LsF can be obtained by taking the square of the load factor.

Consider a discrete version of a load duration curve as shown in Figure 5.14. In this figure, L_p is the peak load, and L_1 , L_2 , and L_3 are the other load levels, which exist for certain time durations as shown in the figure. The total time duration is

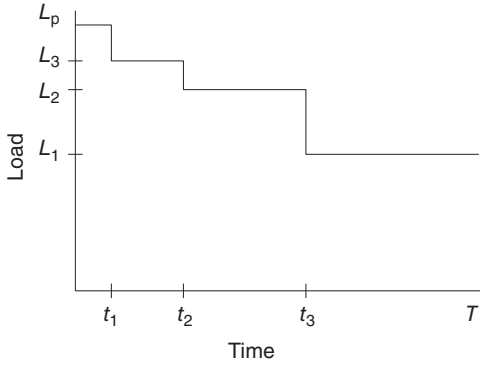


Figure 5.14 Discrete load duration curve.

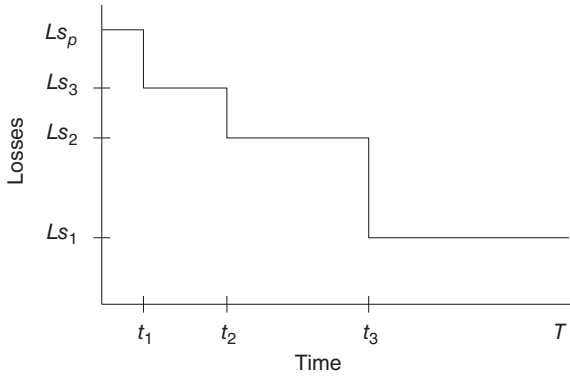


Figure 5.15 Discrete loss duration curve.

T hours. Since resistive losses (which are the predominant losses in distribution systems) at any load are proportional to the square of current, they are proportional to the square of load if we consider voltage to be fixed. Hence, we can obtain loss characteristics for the same duration as shown in Figure 5.15. L_{sp} are the average hourly losses at peak load, and L_{s1} , L_{s2} , and L_{s3} are the losses at other load levels.

From Figure 5.14, we get

$$LF = \frac{L_p t_1 + L_3(t_2 - t_1) + L_2(t_3 - t_2) + L_1(T - t_3)}{TL_p} \quad (5.6)$$

Similarly,

$$LSF = \frac{L_{sp} t_1 + L_{s3}(t_2 - t_1) + L_{s2}(t_3 - t_2) + L_{s1}(T - t_3)}{TL_{sp}} \quad (5.7)$$

For a special case where L_1 , L_2 , and L_3 are equal, we get

$$LF = \frac{t_1}{T} + \frac{L_1(T - t_1)}{L_p T} \quad (5.8)$$

and

$$\text{LsF} = \frac{t_1}{T} + \frac{Ls_1(T - t_1)}{Ls_p T} \quad (5.9)$$

Now, if we consider the peak load of a very short duration or $t_1 \ll T$,

$$\frac{t_1}{T} \rightarrow 0 \text{ and } \frac{(T - t_1)}{T} \rightarrow 1 \quad (5.10)$$

Hence,

$$\text{LF} = \frac{L_1}{L_p} \text{ and } \text{LsF} = \frac{Ls_1}{Ls_p}. \quad (5.11)$$

With $Ls_1 = k(L_1)^2$ and $Ls_p = k(L_p)^2$, we get

$$\text{LsF} = (\text{LF})^2 \quad (5.12)$$

This approximation becomes better as T becomes larger, such as one year, because the peak load typically exists only for an hour or two. Also, the square relationship between the load factor and the LsF is an extreme situation because in reality all the off-peak loads are not equal. Other approximations are also available in the literature. However, for planning purposes, it is acceptable to use this relationship.

5.5 Design Criteria and Standards

In addition to minimizing the cost of the equipment needed for a distribution system, the job of a planning engineer is to ensure that the selected equipment, layout, and installation meet the design criteria and standards. Voltage and service quality, equipment loading, and safety are the three issues relevant to the planning and design of distribution systems. These criteria are either specified by equipment manufacturers and standards organization or set by the utilities for their own use. Sometimes, standards set by utilities are more stringent than those set by manufacturers or the standards organization. Manufacturers specify equipment loading values to prevent the damage to the equipment. Safety is related to providing proper insulation and adequate clearance between equipment. Voltage and service quality affect the customers directly.

5.5.1 Voltage Standards

According to ANSI C84.1 standard, utilities in the United States are required to provide service voltage to customers in the 114–126 V range (on a 120-V nominal). This is Range A, or the preferred range, and is based on $\pm 5\%$ fluctuations around

the nominal. During emergencies, the utilities are allowed to use Range B, which is from 110 to 127 V. With 4 V drop in the building wiring allowed by ANSI C84.1, the customers receive utilization voltage in 110–126 V range under normal operation. The equipment manufacturers use this range to design their equipment. The life of equipment is usually shortened if operated outside of this range. Usually, utilities try to keep the service voltage for all customers at or above 120 V to have a larger safety margin.

Providing voltage within the American National Standards Institute (ANSI)-specified values to all the customers is a big challenge for the planners. The challenge is compounded by the fact that the flow of power over the conductors from the substation to the customers will result in voltage drop, and this voltage drop fluctuates with the fluctuation in load throughout the day. Also, the customers close to the substation see a lower voltage drop compared to the customers away from the substations. In addition to proper system design, utilities use load tap changers, line regulators, capacitors, or a combination of the three to provide service voltage to the customers within the ANSI-specified range.

5.5.2 Conservation Voltage Reduction

It is widely recognized that reducing voltage reduces both the real and the reactive power demand and energy consumption. Therefore, with proper control and monitoring, the utilities can operate the system to provide service voltage in the lower side of the ANSI range, with some customers having service voltage of 114 V, to capitalize on this phenomenon. Several experiments were conducted by various utilities in the 1970s and 1980s to determine the effects of voltage reduction on the system load. Despite some success of a few utilities in implementing voltage control specifically to reduce energy (or *conservation voltage reduction* (CVR)) and peak power, such control presently is practiced only by a few utilities. Uncertainties in the level of load reduction and fear of voltage-related complaints led to the lack of interest among utilities on this issue. Interest in this topic has grown lately due to national interest in energy conservation and concerns about environmental impacts and to efforts to reduce carbon emissions. Implementation of voltage control for energy and power reduction allows utilities to participate in energy conservation efforts, which makes them good corporate citizens and promotes their “green image.”

Various studies conducted on this issue show that real load-to-voltage sensitivities (% change in real power to 1% change in voltage) may range from 0.4 to as high as 2.5, but the average is in the 0.8–1.0 range. The reactive load-to-voltage sensitivity is larger with a range of 3–5 with an average of about 4. The energy sensitivity reported for a period of 24 hours ranges from 0.6 to 1.237. However, for a given system, the energy-to-voltage sensitivity is usually lower than the real load-to-voltage

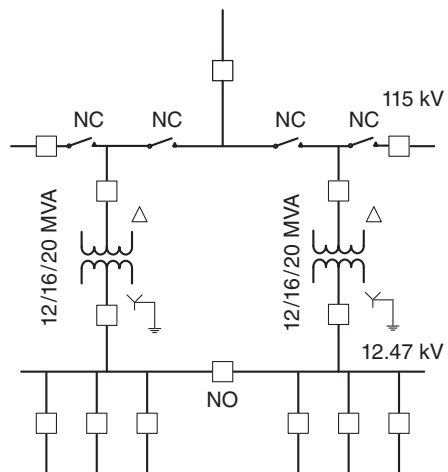
sensitivity. The exact load-to-voltage sensitivity for any substation/feeder will be impossible to predict and will vary each day, throughout the day, and throughout the year. The commonly acceptable number by industry for real load-to-voltage sensitivity is 1.0, and for reactive load-to-voltage sensitivity is 3.0. The general practice of utilities regarding voltage reduction is to either use it on a continuous basis to reduce energy or only during emergency to reduce power, or a combination of both.

5.6 Distribution System Design

5.6.1 Substation Design

Distribution substation is where transmission or subtransmission system ends and distribution system begins. Power comes into distribution substation between 34.5 and 230 kV range, and it is reduced to a voltage level between 2.4 and 34.5 kV. The voltage level at the distribution level is determined by the load density and the distance to which power must be delivered from the substation. Usually, the substations are located to uniformly cover the whole service area without having excessively long distances between some customers and the substation. However, this rule may not apply in rural settings. The layout of an example substation is shown in Figure 5.16. In this figure, there are three 115-kV incoming lines, and they are connected to the 115-kV bus. The 115-kV bus is normally operated as one contiguous bus with four normally closed (NC) switches. In emergencies, some of these switches can be opened to operate this bus in a different configuration. Note that each incoming line has its own circuit breaker. This is the simplest

Figure 5.16 Typical layout of a substation for a semiurban area (NC, normally closed and NO, normally open).



bus arrangement. Other arrangements, such as main and transfer bus, ring bus, and breaker-and-a-half bus schemes, can be used to enhance reliability. However, these schemes are more expensive, and some of them also complicate the protection system.

Distribution substations can have one to three transformers. The substation of Figure 5.16 has two transformers, each rated 12/16/20 MVA with 115 kV Δ /12.47 kV Y-grounded connections. It is preferred to have both transformers of the same capacity, which is selected based on the total load in the service territory of the substation. The tie breaker on the 12.47-kV bus is normally open to avoid parallel operation of transformers, but can be closed under emergencies. Parallel operation of transformers is not recommended because that creates circulating currents and increases losses. Typically, the transformers are loaded to 50% of their highest rating during peak load conditions. This is done so that if one of the transformers fails, the other one is able to pick up the whole load. Thus, if the substation has three transformers, they can be loaded to 67% of their highest rating. In this case, the other two transformers will pick up the load of the failed transformer. Several transformers are needed in the substation to further increase the loading on transformers. If the substation has only one transformer, its failure will result in sustained loss of service to the customers. This was the old way to select sizes of distribution substation transformers. In the new approach, the distribution systems can be designed with automation to receive support from other substations in case of failures. Automation allows quick operation of switches and thus load transfer to other transformers. The distribution substation transformers usually have Δ /Y configuration to provide neutral on the low-voltage side for distribution of single-phase power to customers using one of the phases and the neutral. The rule of keeping Y-connection on the high-voltage side of the Δ /Y transformers is overruled in this case to make the neutral available on the low-voltage side.

The low-voltage side of the transformers are connected to the 12.47-kV bus, which is operated in two segments with the normally open (NO) switch in between. Closing the switch would parallel the two transformers, which is not desirable. Each segment of the low-voltage bus has a number of feeders, which can range from 1 to 4 per segment. Deciding on how many feeders to use is a subjective process, and it depends on the load to be served and the capacity of the transformer. A typical feeder is designed to carry 4–8 MVA. In the substation shown in Figure 5.16, there are three feeders per transformer, with a total of six feeders.

5.6.2 Design of Primary Feeders

Primary feeders carry power from substations to customers and thus are the backbone of the distribution systems. Except for a very densely populated service area,

Figure 5.17 Single-feeder layout of distribution feeders.

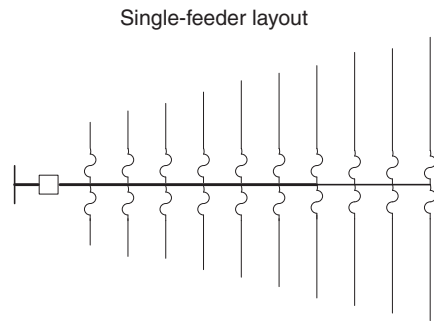
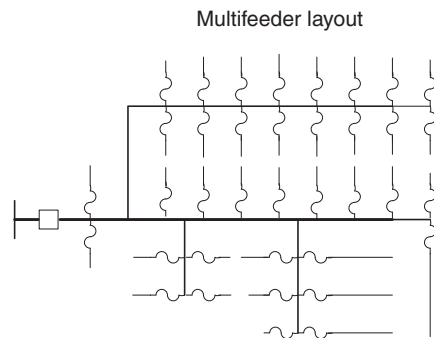


Figure 5.18 Multifeeder layouts of distribution feeders.



such as downtowns of large cities, distribution systems have a radial structure. In a radial structure, the power from the utility to customers flows only in one direction. Although the radial structures provide lower reliability, they are substantially cheaper to build and much easier to operate and protect. Therefore, distribution systems are predominantly radial with the feeders arranged in a tree-like structure. The root of this tree is at the substation. From this root the main three-phase branch of the primary feeder or the trunk comes out. This primary feeder may split into multiple subfeeders or remain as one throughout its service territory. Examples of single and multifeeder layouts are shown in Figures 5.17 and 5.18, respectively. Due to the radial nature, the load on the primary feeders reduces as we move away from the substation. Therefore, in some cases, conductor sizes for the primary feeders are tapered as we move away from the substation.

Each primary feeder leaving a substation is assigned a part of the service area of the substation so that each primary feeder has roughly equal service areas. In an ideal situation, a substation with four feeders will have the division of the service area as shown in Figure 5.19. Similarly, a substation with six feeders will have the service area division as shown in Figure 5.20. In real life such clean demarcation between service areas is not possible, but the planners try to maintain this division as much as possible. If a substation has a higher number of primary feeders, the service territory is divided equally.

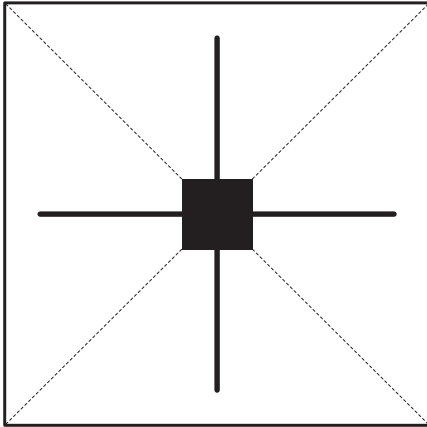


Figure 5.19 Service territory division for four primary feeders.

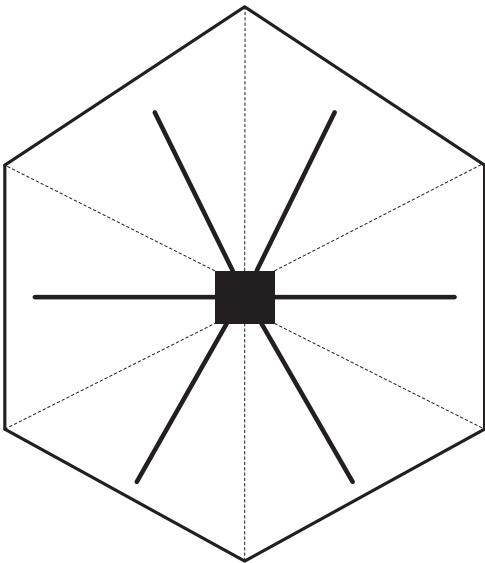


Figure 5.20 Service territory division for six primary feeders.

Although distribution systems operate in a radial configuration, most of the primary main feeders have a NO (Normally Open) switch or circuit breaker, called the tie switch, at their ends. When a permanent fault occurs on a primary feeder so that it loses connection to the substation, the device at the end of the feeder is closed to restore power to some of the customers receiving power from the faulted feeder. The feeder on the other side of this device could be connected to the same transformer, or to a different transformer from the same substation or to a transformer from a different substation. Each of them provides a different level of reliability and has different costs associated with it. Therefore, careful planning of the feeder

layout is needed to achieve the desired results. For example, if two feeders are connected to the same transformer, the loss of that transformer will result in no backup possibilities for customers connected to these feeders. Power will have to be restored at the substation by connecting these feeders to a different transformer. On the other hand, if they are connected to different transformers, the loss of a transformer will result in power loss to only one of the feeders. Power to this feeder can be supplied by connecting to the other feeder by closing the tie switch.

Connected to the primary main feeders are the primary lateral feeders, which are usually single phase, with phase and neutral conductors. They carry power from the primary main to customers. Primary main feeders are typically along the city streets, but the lateral feeders are mostly in the utility easement behind houses in residential subdivisions in cities and towns. The phase selection (A, B, C) for specific laterals branching off from the primary main is done to maintain a rough balance among the three phases. Figure 5.21 shows the feeders and streets in a typical city in the United States.

5.6.3 Design of Secondary Systems

The secondary system is that part of the distribution system which is closest to the customers. It includes the distribution transformers, secondary feeders, and service drops. The secondary system is single phase for residential customers, but it could be three phase for large industrial and commercial customers.

Distribution transformers are small transformers, ranging in size from 5 to 100 kVA with 120 V/240 V secondary. The size of the transformer used in a specific location depends on the number and the class of customers served by it. Typically, a transformer serves four to eight medium-sized single-family homes. If homes are very large, a transformer may serve only one or two homes. On the other hand, a single transformer may serve several apartments in a complex. Typical layouts of secondary systems are shown in Figure 5.22.

5.6.4 Underground Distribution Systems

A large percentage of distribution systems all over the world is overhead. Cost is the main factor for this because the cost of an underground system is 5–10 times more than the overhead system. However, in densely populated areas and in business districts, the distribution systems are underground. In other areas, aesthetics is the motivation for underground distribution systems. Many cities are mandating underground systems within new housing developments. Even if cities do not mandate, some developers use underground distribution in high-cost subdivisions. In these situations, laterals and secondary systems are underground, but the primary main is still overhead.

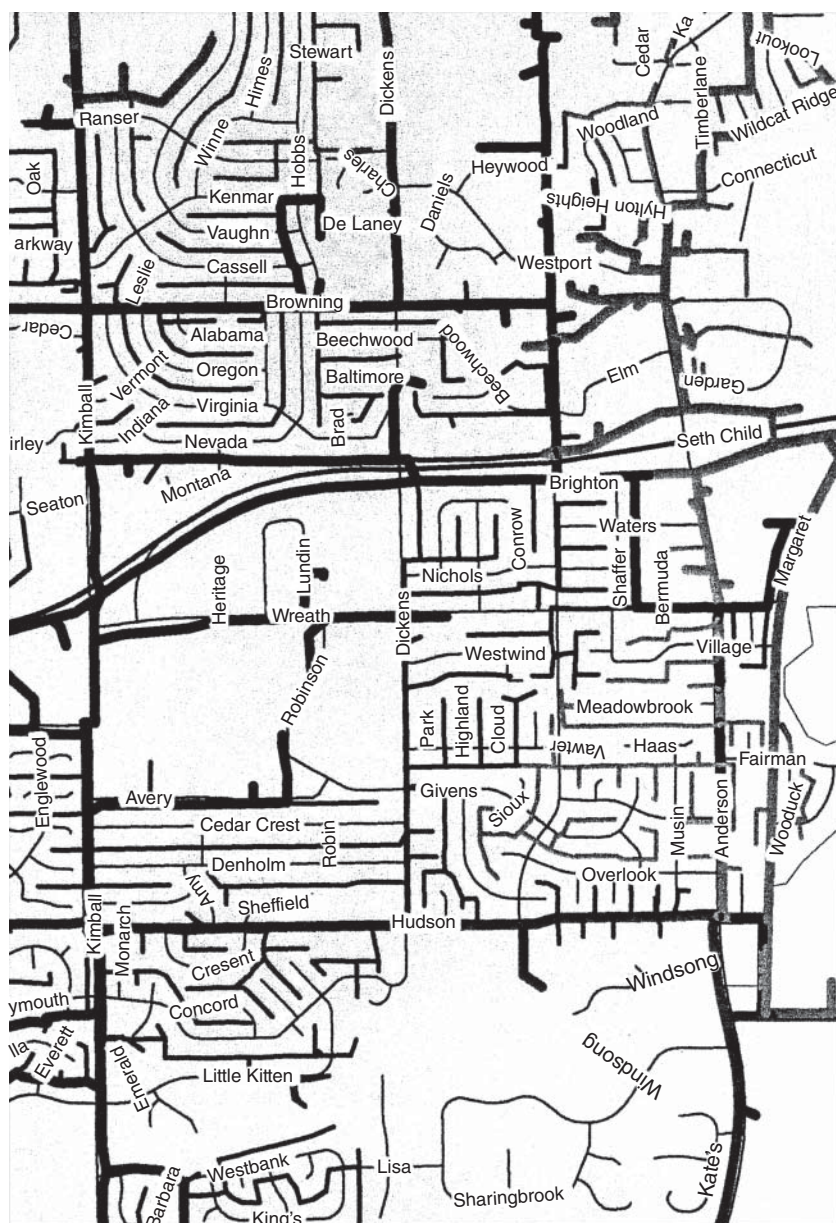


Figure 5.21 Feeders along with the roads in a typical distribution system in the United States.

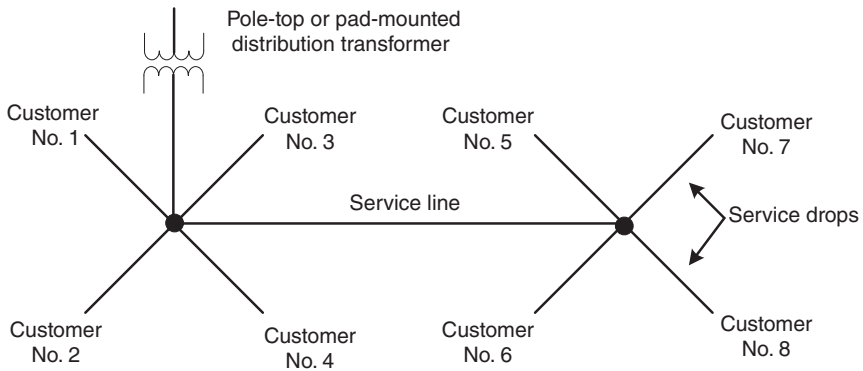


Figure 5.22 Illustration of a secondary system for service to eight customers.

Overhead systems are more exposed to the environment and therefore more prone to failure. Several studies have shown that more than 50% of the outages on overhead distribution systems are caused by environmental factors including weather, trees, and animals. One advantage of overhead systems is that the location of faults and repairs are relatively easy. In contrast, fewer faults occur in underground systems; but if a fault does take place, it is very difficult and time consuming to locate that fault. The repairs are more difficult and take a longer time.

5.6.5 Rural vs. Urban Systems

In general, the distribution systems can be divided into three types based on the load density: urban, semiurban, and rural. Urban systems have very high load density and are designed with lower substation spacing, higher size substation transformers, and bigger conductors for feeders. Since the substation spacing is small, the total distance from the substation to the farthest customer served from the substation is not very long. Each substation also has two or more power transformers. Rural systems have very low load density and the loads are spread out. Therefore, the substation spacing is very large, the transformer sizes are small, and the conductors for the feeders are relatively small in size. Further, the substation may have only one transformer. Generally, sufficient number of NO tie points are available in the urban systems, but very few are available in the rural systems. The tie points improve reliability of the system significantly and are easy to locate in urban systems. It is very difficult and expensive to locate tie points in rural systems, and also, as the number of customers in rural systems is low, the impact on reliability is only marginal. As a consequence of this, the reliability of urban systems is substantially higher than the rural systems. The semiurban systems are in between these two systems in all respects.

In urban systems, since the total load served by feeders is higher, and feeders are shorter in length, by design, and they are mostly thermally limited. That is, the total load that a substation can supply is based on the thermal loading capacity of the transformers and the feeders. Voltage drop on the feeders is usually not a problem. The rural systems, on the other hand, are voltage-drop limited and that is due to the length of the feeders, which makes the voltage drops on them are quite large. Hence, they can only carry a limited amount of load to ensure that the last customer on the feeder gets service voltage within the limits. The semiurban systems are in between. They may have both problems, depending on the size and type of equipment selected while designing the system.

5.7 Cold Load Pickup (CLPU)

5.7.1 CLPU Fundamentals

Loads on a predominantly residential feeder can be divided into two categories those that are thermostatically controlled and those that are manually controlled. Thermostatically controlled devices such as air conditioners, heaters, heat pumps, and refrigerators provide the largest contribution to the total load in a typical house. Manually controlled loads are switched on and off by occupants of the house in a random fashion based on their need. The aggregate load of a group of houses is less than the connected load under normal operation due to diversity among loads. However, after an extended interruption, some or all thermostatically controlled devices would be on as soon as the power is restored. Similarly, the aggregate load of manually controlled devices immediately after restoration would be higher than normal because more people may want to use different devices. The lifestyle of the occupants of the house has a significant influence on the contribution of these loads to the total load of the house.

The load phenomenon that is experienced upon restoration after an extended power interruption is called cold load pickup (CLPU). CLPU can be categorized into four phases according to the current levels and durations. These phases are inrush, motor starting, motor running, and enduring phases. The first three phases last approximately less than 15 s, and the current may reach 5–15 times of the preoutage current. The enduring phase follows the third phase and continues until the normal diversity among the loads is reestablished. The load in this phase may vary from two to five times of the diversified load level, and this phase may last for several hours, depending on the outage time, outside temperature, and the type and ratings of devices. Several utilities also experienced this phenomenon during implementation of direct load control on thermostatically controlled devices.

CLPU appeared first in the literature in the 1940s as a problem related to high inrush currents that last a few seconds and prevent the circuit from being reenergized after extended outages [3, 4]. Application of very inverse characteristic relays and sectionalizing the distribution systems were some of the solutions engineers used to overcome this problem. Since then, penetration of thermostatically controlled devices in distribution systems has increased. These types of loads may cause restoration problems during CLPU before they cause serious overload problems in normal operation. Therefore, sustained or enduring load after restoration becomes an important issue for loading limitations of distribution equipment.

5.7.2 CLPU Models

With increasing penetration of thermostatically controlled loads, the enduring components of CLPU grabbed the attention of researchers in the 1970s and 1980s. Some experimental studies were conducted to predict the magnitude and the duration of the peak demand upon service restoration following a long power outage [5]. Substantial work on physically-based modeling of this phenomenon of loads took place [6]. Some of it was initiated under the sponsorship of the U.S. Department of Energy. The main idea is to develop a physically-based load model for individual loads as a function of weather and human use patterns and then use aggregation of these loads to find the total demand. Aggregated load behavior, when a large number of customers are considered, can be determined based on these models either by use of numerical techniques to solve partial differential equations derived from individual load model or by Monte Carlo simulation based on the stochastic difference equations. Some simplified models have also been used to find the effect of CLPU on substation and distribution transformers [7]. Because the thermal response of a transformer to a load is slow, simplified models are sufficient to analyze its loading capabilities. Simplified models in which postoutage load is constant for some time and then decreases linearly and piece-wise linear CLPU models have been used to find the overloading capability of transformers.

Figure 5.23 shows an example of data recorded by a utility in the east coast of the United States after restoration following an interruption of 1 h and 39 min around midnight during winter. The load prior to interruption was 749 kW. The graph clearly shows that the load in the first 15-min period immediately after restoration is 1451 kW, which is approximately two times the load prior to interruption. Also, the load returns to normal in the third interval. The value of the initial load after restoration and the time taken to return to normal would depend on the load characteristics, duration of interruption, time of interruption, and ambient temperature. Since the data was recorded for 15-min intervals, it is hard to say more

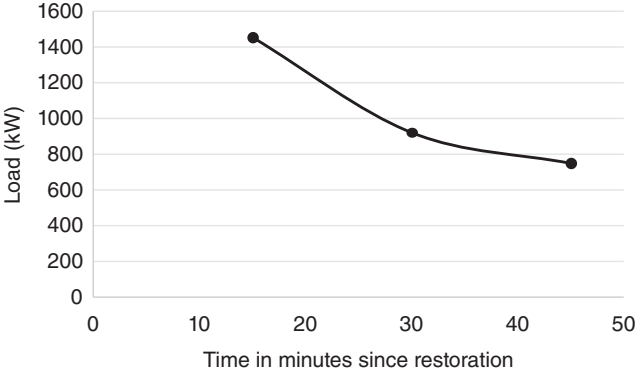


Figure 5.23 Load upon restoration following a long outage during winter recorded by an east-coast utility in the United States. The dots show the average load for 15 min prior to that time.

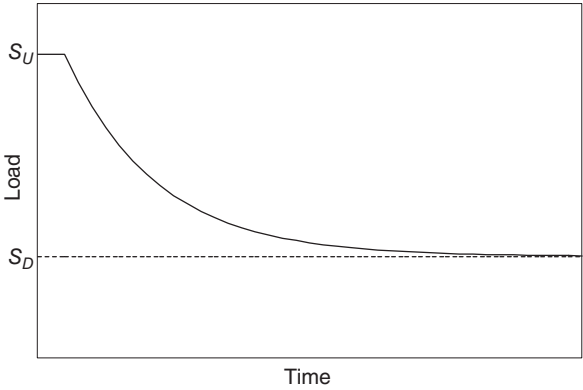


Figure 5.24 Delayed exponential model for cold load pickup. S_U is the undiversified load, and S_D is the diversified load.

about the cold pickup characteristics based on this graph. Data recorded on 5- or 1-min resolution could provide more information about CLPU.

Simulations as well as experimental data suggest that aggregate load on a residential feeder in distribution system during CLPU can be represented by a delayed exponential model [7], as shown in Figure 5.24.

5.7.3 Impacts of CLPU

CLPU can impact both the operation and design of distribution systems. These impacts are dependent on the type of system. Since load during CPLU is much higher than normal load, it causes overloading of the transformer and conductors and excessive voltage drops on the feeders. The first factor is obviously more crucial for the urban systems, whereas the second factor is more crucial for the rural

systems. In either case, it may not be possible to restore the entire load in one step. One possible approach is to install remotely operated sectionalizing switches on the main feeder to split the system into several sections. During restoration, the sections can be restored in steps. Load behavior of sections and restoration sequence of these sections play an important role in the restoration procedure. Based on the load dynamics of each section, the restoration procedure should be chosen in such a way that the selected restoration objectives are met while the loading of transformers and feeders is within limits, and there are no voltage-drop violations. For example, one of these objectives is to minimize customer interruption duration because it directly affects the reliability of the system. The shorter the customer interruption is, the more reliable the system will be. We will discuss additional details on CLPU when we revisit it in a later chapter.

5.7.4 Operating Limits

According to ANSI/IEEE PC57.91-1994 standard, the load on a transformer can exceed the rated load under an emergency situation, but the loss of life of the transformer should not exceed 4% [8]. Further, the maximum top-oil temperature should not exceed 110 °C, the maximum hottest spot winding temperature should not exceed 180 °C, and the maximum short time loading should not be more than two times the maximum normal rating. Similarly, short time emergency loading of conductors is 133% of the normal full load.

At all times, the supply voltage to all the customers must be within the limits specified by ANSI C84.1. Of the total voltage drop from the substation to the service entrance, approximately 2.5% under normal operation and 3.54% under emergency operation can be assumed for the laterals, and the rest for the main feeder. Hence, if the voltage at the substation is considered to be at the upper end of the zone, the total permissible voltage drop on the main feeders would be 7.5% under normal operation and 10.62% under emergencies. Thus, during restoration under CLPU conditions, the voltage drop should not exceed 10.62%.

5.8 Asset Management

Asset management, as the name implies, is an approach for utilization of assets in an optimal way. This applies both for planning and operation. During the planning process, decisions have to be made on replacing or updating the existing equipment while keeping in mind the age and current operating status of these equipment. Waiting for too long can have disastrous results, but at the same time replacing equipment too early will cost utilities more money. Therefore, decisions have to be made while accounting for risk associated with selected

actions. Similarly, during operation, real-time decisions on loading of equipment have to be made based on their conditions and current operating status. Sustained overloading could result in equipment failure, but on the other hand, without overloading service to some customers would have to be curtailed. Hence, risk of overloading has to be balanced with loss of power to some customers.

The issue of asset management has come into limelight because a large part of the existing distribution and transmission infrastructure is very old, with some equipment with age higher than 40 years. Complete replacement of these equipment is cost prohibitive. So a planned replacement strategy is required.

Problems

- 5.1** Power demands of a house recorded for a house on a 15-min basis for a duration of two hours are given in Table 5.3.

Table 5.3 Data for Problem 5.1.

Period	Power demand (kW)
1	1.2
2	1.4
3	2.0
4	1.8
5	1.6
6	1.4
7	1.2
8	1.0

Find the average demand for each of the two hours and for the total duration of two hours.

- What is the total energy consumed in the first hour?
 - What is the total energy consumed in the total duration of two hours?
- 5.2** Energy consumptions in a house recorded on five-minute basis for a one-hour duration are shown in Table 5.4.
- Find the power demand for each period.
 - What is the average power demand for the hour?
 - What is the total energy consumption for the hour?

Table 5.4 Data for Problem 5.2.

Period	Energy consumed (kWh)
1	0.5
2	0.6
3	0.8
4	1.0
5	0.9
6	0.8
7	1.0
8	0.8
9	0.7
10	0.6
11	0.4
12	0.5

5.3 Energy consumptions recorded on a one-hour basis for four hours for five houses are shown in Table 5.5.

Table 5.5 Energy consumption data of houses for Problem 5.3.

Hour	Energy consumption (kWh)				
	House number				
	1	2	3	4	5
1	2.0	4.8	3.2	1.2	1.6
2	2.4	1.6	1.2	3.0	2.0
3	3.2	2.0	1.6	1.0	0.8
4	4.0	0.8	2.0	2.2	3.0

- (a) Find the combined power demand of five houses for each hour.
- (b) What is the coincidence factor for these houses?
- (c) What is the diversity factor?

5.4 The hourly load power demands for a primary distribution feeder of a power company for a typical winter day are given in Table 5.6.

Table 5.6 Data for Problem 5.4.

Hour	Load (kW)		
	Street lighting	Residential	Commercial
1	200	200	50
2	200	200	50
3	200	200	50
4	200	200	50
5	200	200	50
6	180	300	50
7	150	400	50
8	0	600	50
9	0	500	200
10	0	500	600
11	0	500	600
12	0	500	600
13	0	500	600
14	0	500	600
15	0	500	600
16	0	500	600
17	100	600	600
18	200	800	600
19	200	800	600
20	200	600	600
21	200	600	600
22	200	300	300
23	200	300	300
24	200	300	300

- (a) Find the DFs for each load type with the total connected loads of 200, 1200, and 800 kW for street lighting, residential, and commercial loads, respectively.
- (b) Find the diversified maximum demand and the coincidence factor of the feeder.
- (c) Find the total combined energy consumption in kWh for the day. Use this information to find the load factor of the feeder for this day.

- (d) If the losses on this feeder at peak load are 5% of the peak load, find the average hourly loss using LsF.
- (e) Find the total cost due to losses on this day with an energy cost of 3 cents/kWh.

5.5 Table 5.7 shows the hourly load on a substation transformer for a 24-h duration.

Table 5.7 Data for Problem 5.5.

Hour	Load (MW)	Hour	Load (MW)	Hour	Load (MW)	Hour	Load (MW)
1	4.1	7	3.5	13	10.4	19	17.8
2	3.5	8	5.1	14	12.1	20	15.1
3	3.1	9	6.2	15	14.2	21	12.8
4	2.3	10	7.4	16	15.9	22	8.8
5	2.6	11	8.9	17	17.8	23	6.5
6	3.2	12	9.8	18	20.2	24	5.2

- (a) Draw a plot showing load in MW on y-axis as a function of time.
- (b) Draw a load duration curve for the given data with a step size of 1 MW.
- (c) Draw a load duration curve for the given data with a step size of 0.5 MW.
- (d) What is the load factor for parts (b) and (c)?

5.6 A 100 sq. mile (10×10 miles) area has a load density of 1.5 MW/square mile with 250 customers per square mile. The load power factor is 0.9 lagging. Each primary feeder can carry load in 4–8 MVA range, and the maximum length allowed for the feeders is six miles. Find the substation spacing, number of substations, substation capacity, number of transformers needed per substation and their capacity, and the number of primary feeders per substation. (Note: This is a challenge problem and does not have a unique solution.)

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